

Data Cleaning Log

HeavySuppliersWarehouseDatasets

Scope

This log covers every change made to the 12 original CSV files during cleaning, every check run to verify data quality, and every issue found along the way — including which ones were fixed, which were left alone on purpose, and why.

Starting point: 12 files, 605,141 rows, 134 columns. Profiling (see separate profiling report) had already confirmed 0 exact duplicate rows and 0 orphaned foreign keys across 17 tested relationships before any cleaning began.

1. Changes applied to all files

1.1 Date columns converted to proper datetime type

The following columns were being read in as plain text and were converted to actual dates:

- invoices.csv: invoice_date, due_date
- payments.csv: payment_date
- sales_orders_header.csv: order_date, delivery_date
- purchase_orders_header.csv: order_date, expected_delivery_date, received_date
- stock_ledger.csv: movement_date
- customers.csv: customer_since, last_purchase_date
- products.csv: last_purchase_date

1.2 branches.csv — warehouse_capacity split

warehouse_capacity was stored as text with a unit suffix (e.g. "45230 sqft"). The number was extracted into a new column, warehouse_capacity_sqft, and the original text column was dropped.

2. Issues found and left unchanged (documented, not fixed)

2.1 inventory_master.csv — stock threshold scale mismatch

current_stock values (roughly 88,000–121,000) are on a completely different scale from max_stock, reorder_level, and safety_stock (tens to hundreds). Verified current_stock against the latest running_balance in stock_ledger.csv — matches exactly on all 180 rows, confirming current_stock itself is correct. The threshold fields appear to have been set on an unrelated scale.

Action taken: No values were changed. Anyone building stockout, reorder, or overstock logic must account for this.

2.2 purchase_orders_header.csv — received_date nulls

received_date is null for all 2,370 rows where po_status = Cancelled, and filled for every one of the 21,630 Received rows — zero exceptions either way. This is expected (a cancelled order was never received), not missing data.

Action taken: No values were changed.

3. Issues found and fixed

3.1 invoices.csv / payments.csv — non-unique primary keys

invoice_id and payment_id are meant to be unique per row but weren't:

- 197 invoice_id values reused across 393 rows — 196 of those pairs have genuinely different data (different customer, order, amount), not simple duplicate rows (0 exact duplicate rows were found in either file).
- 202 payment_id values reused across 403 rows — 201 genuine collisions, same pattern.

Confirmed present in the original raw files before any cleaning touched them — this is a source data defect, not something introduced during cleaning.

Fix applied: Added a new, guaranteed-unique surrogate key to each file — invoice_uid and payment_uid (row-based IDs, e.g. INVUID-000001). These do not derive from the original ID columns, so they cannot collide. The original invoice_id / payment_id columns were left completely untouched for reference.

Downstream consequence found and flagged, not fixed: because payments.csv links to invoices only via the (non-unique) invoice_id, 417 payment rows and 393 invoice rows cannot be confidently matched to a single invoice — there is no way to tell from the data which of two colliding invoices a given payment belongs to. Rather than guess, a new boolean column, invoice_id_ambiguous, was added to both files, True on every affected row. This surfaced as two symptoms during validation before the root cause was traced:

- 247 Paid invoices where summed payments didn't match grand_total
- 35 Unpaid invoices that had a payment record attached

Both symptoms are fully explained by the ID collision (100% overlap with the flagged rows) — they are not separate defects.

3.2 products.csv — one incorrect margin_percentage

P002 (Engine Oil Filter OF-90) listed margin_percentage = 37.3, but its own unit_cost (450) and unit_price (720) calculate to 60.0%. Corrected to 60.0.

Three other products (P001, P005, P014) showed a 0.1 percentage-point difference between their stated margin and the recalculated value — this is normal rounding, not an error, and was left unchanged.

4. Validation checks run (no issues found)

- Primary/composite key uniqueness checked on all 12 tables: branches, customers, suppliers, products, purchase_orders_header, sales_orders_header, stock_ledger (single-column keys), plus inventory_master (product_id+branch_id), purchase_orders_lines (po_id+line_number), sales_orders_lines (so_id+line_number) — all clean except invoices/payments (see 3.1).
- customer_rating and reliability_score within expected 1–5 range — no violations.
- No customer over their credit_limit.
- No negative or zero values in quantity, cost, price, or payment fields.
- No backwards date ranges (order date after delivery date, etc.).
- Header-level tax math (total + GST = grand_total) consistent across invoices, sales_orders_header, and purchase_orders_header — no mismatches.

5. Files affected by this cleaning pass

File	Changed?	What changed
branches.csv	Yes	Dates n/a; warehouse_capacity split into warehouse_capacity_sqft
customers.csv	Yes	Date columns converted
suppliers.csv	No	—

products.csv	Yes	Date column converted; P002 margin corrected
inventory_master.csv	No	Scale mismatch documented, not altered
purchase_orders_header.csv	Yes	Date columns converted
purchase_orders_lines.csv	No	—
sales_orders_header.csv	Yes	Date columns converted
sales_orders_lines.csv	No	—
invoices.csv	Yes	Date columns converted; invoice_uid and invoice_id_ambiguous added
payments.csv	Yes	Date column converted; payment_uid and invoice_id_ambiguous added
stock_ledger.csv	Yes	Date column converted